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Exit device coordination mechanisms

Abstract

An exemplary exit device includes a pushbar assembly, a remote latching assembly, and a coordination mechanism. The pushbar assembly includes a latch control assembly, and the coordination mechanism biases the latch control assembly toward its actuated state. The remote latching assembly includes first and second latch mechanisms, and the latch control assembly is operable to actuate the latch mechanisms. Each of the latch mechanisms at least selectively urges the latch control assembly toward its deactuated state such that the remote latching assembly exerts a variable deactuating force on the latch control assembly. The coordination mechanism selectively retains the latch control assembly in its actuated state, thereby selectively retaining at least one of the latch mechanisms in a corresponding actuated state. When the deactuating force exceeds a threshold force value, the latch control assembly and the latch mechanisms return to the deactuated states thereof.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The current application is a continuation of U.S. patent application Ser. No. 16/042,214 filed Jul. 23, 2018 which claims the benefit of U.S. Provisional Patent Application No. 62/671,518 filed May 15, 2018, the contents of each application are incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure generally relates to exit devices, and more particularly but not exclusively relates to exit devices including one or more remote latching mechanisms.

BACKGROUND

(2) Exit devices are commonly installed to doors to provide for rapid egress, and typically include one or more latching mechanisms and a pushbar assembly. Each latching mechanism is operable to engage a door frame to retain the door in a closed position, and the pushbar assembly is operable to retract or actuate the latching mechanisms to permit opening of the door. Certain exit devices include a remote latching assembly in which one or more of the latching mechanisms is positioned remotely from the pushbar assembly. Such remote latching assemblies typically include a top latch mechanism mounted at the top of the door, and often further include a bottom latch mechanism mounted at the bottom of the door. The latch mechanisms are typically connected to the pushbar assembly via a pair of connectors, such as rigid rods or flexible cables. Certain remote latching assemblies are mounted to the surface of the door, while others are concealed within a set of cavities and channels that are formed in the door.

(3) One difficulty associated with remote latching assemblies of this type is that premature extension of the bottom bolt may cause the bolt to drag along the floor during movement of the door. To address this issue, certain existing exit devices include mechanisms that retain the bottom bolt in its retracted position until the door reaches its closed position. When the connectors are provided in the form of rods, the rigidity of the rods enables the top latch mechanism to exert a pushing force that can be used to retain the lower rod in its raised position, such as via a lever or rack and pinion assembly. Such rods are not without their drawbacks, however. For example, when used in a concealed remote latching assembly, rods typically require much larger cutouts than would be required by cables. Larger cutouts can reduce the structural integrity of the door, particularly in wood doors.

(4) While flexible pull cables typically allow for smaller cutouts, their inability to transmit pushing forces complicates the selective retention of the bottom bolt in its retracted position. While certain existing remote latching assemblies include retention mechanisms that provide the desired functionality, these retention mechanisms typically require that the door be provided with a mortise cutout, which can also reduce the structural integrity of the door. Furthermore, the incorporation of such retention mechanisms in surface-mounted remote latching assemblies is hindered by the fact that these retention mechanisms typically cannot be mounted within the pushbar assembly itself. As such, certain surface-mounted remote latching mechanisms are restricted to the use of rods.

(5) In light of the foregoing, it is evident that many existing exit devices suffer from certain limitations, such as those relating to selectively preventing extension of the bottom bolt while maintaining structural integrity of the door. For these reasons among others, there remains a need for further improvements in this technological field.

SUMMARY

(6) An exemplary exit device includes a pushbar assembly, a remote latching assembly, and a coordination mechanism. The pushbar assembly includes a latch control assembly, and the coordination mechanism biases the latch control assembly toward its actuated state. The remote latching assembly includes first and second latch mechanisms, and the latch control assembly is

operable to actuate the latch mechanisms. Each of the latch mechanisms at least selectively urges the latch control assembly toward its deactuated state such that the remote latching assembly exerts a variable deactuating force on the latch control assembly. The coordination mechanism selectively retains the latch control assembly in its actuated state, thereby selectively retaining at least one of the latch mechanisms in a corresponding actuated state. When the deactuating force exceeds a threshold force value, the latch control assembly and the latch mechanisms return to the deactuated states thereof. Further embodiments, forms, features, and aspects of the present application shall become apparent from the description and figures provided herewith.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a perspective illustration of a closure assembly including a door and an exit device according to certain embodiments.
- (2) FIG. 2 is a cross-sectional illustration of a pushbar assembly according to certain embodiments.
- (3) FIG. 3 is a partially-exploded assembly view of a portion of the closure assembly illustrated in FIG. 1.
- (4) FIG. 4 is a perspective illustration of a top latch mechanism that may be utilized in connection with certain embodiments.
- (5) FIG. 5 is a perspective illustration of a bottom bolt mechanism that may be utilized in connection with certain embodiments.
- (6) FIG. 6 is a perspective illustration of a coordination mechanism according to certain embodiments.
- (7) FIG. 7 is a cross-sectional illustration of the coordination mechanism illustrated in FIG. 6.
- (8) FIG. 8 is a plan view of a portion of the pushbar assembly illustrated in FIG. 2.
- (9) FIGS. 9a through 9c respectively illustrate a portion of the exit device illustrated in FIG. 1 with the closure assembly in a secured condition, an unsecured condition, and an open condition.
- (10) FIG. 10 is a perspective illustration of a coordination mechanism according to certain embodiments.
- (11) FIG. 11 is an exploded assembly illustration of a coupling assembly according to certain embodiments.
- (12) FIG. 12 is a perspective illustration of a portion of an exit device having the coupling assembly installed thereto.
- (13) FIG. 13 is a schematic flow diagram of a process according to certain embodiments.
- (14) FIG. 14 illustrates a portion of an exit device during one stage of the process illustrated in FIG. 13.
- (15) FIG. 15 is an exploded assembly view of a coordination mechanism according to certain embodiments.
- (16) FIG. 16 illustrates a portion of an exit device having the coordination mechanism of FIG. 15 installed thereto.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

- (17) Although the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.
- (18) References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature,

structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. It should further be appreciated that although reference to a “preferred” component or feature may indicate the desirability of a particular component or feature with respect to an embodiment, the disclosure is not so limiting with respect to other embodiments, which may omit such a component or feature. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to implement such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(19) Additionally, it should be appreciated that items included in a list in the form of “at least one of A, B, and C” can mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C). Further, with respect to the claims, the use of words and phrases such as “a,” “an,” “at least one,” and/or “at least one portion” should not be interpreted so as to be limiting to only one such element unless specifically stated to the contrary, and the use of phrases such as “at least a portion” and/or “a portion” should be interpreted as encompassing both embodiments including only a portion of such element and embodiments including the entirety of such element unless specifically stated to the contrary.

(20) In the drawings appended hereto, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures unless indicated to the contrary. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

(21) As used herein, the terms “longitudinal,” “lateral,” and “transverse” are used to denote motion or spacing along three mutually perpendicular axes, wherein each axis defines two opposite directions. In the coordinate system illustrated in FIG. 1, the X-axis defines first and second longitudinal directions, the Y-axis defines first and second lateral directions, and the Z-axis defines first and second transverse directions. Additionally, the descriptions that follow may refer to the directions defined by the axes with specific reference to the orientations illustrated in the Figures. More specifically, the longitudinal (X) directions may be referred to as “proximal” (X.sup.+) and “distal” (X.sup.-), the lateral (Y) directions may be referred to as “upward” (Y.sup.+) and “downward” (Y.sup.-), and the transverse (Z) directions may be referred to as “forward” (Z.sup.+) and “rearward” (Z.sup.-). These terms are used for ease and convenience of description, and are without regard to the orientation of the system with respect to the environment. For example, descriptions that reference a longitudinal direction may be equally applicable to a vertical direction, a horizontal direction, or an off-axis orientation with respect to the environment.

(22) Furthermore, motion or spacing along a direction defined by one of the axes need not preclude motion or spacing along a direction defined by another of the axes. For example, elements which are described as being “laterally offset” from one another may also be offset in the longitudinal and/or transverse directions, or may be aligned in the longitudinal and/or transverse directions. The terms are therefore not to be construed as limiting the scope of the subject matter described herein.

(23) With reference to FIG. 1, illustrated therein is a closure assembly **60** including a swinging door **70** and an exit device **90** mounted to the door **70**. The door **70** is mounted to a doorframe for swinging movement between an open position and a closed position, and the exit device **90** is configured to selectively retain the door **70** in the closed position. In certain embodiments, the closure assembly **60** may be considered to further include the doorframe. As described herein, the closure assembly **60** has a plurality of states or conditions, including a secured condition, an unsecured condition, and an open condition. In the secured condition, the door **70** is in its closed

position, the exit device **90** is in a deactuated state, and the exit device **90** engages the doorframe and retains the door **70** in its closed position. Actuation of the exit device **90** causes the closure assembly **60** to transition to the unsecured condition, in which the door **70** is capable of being moved from its closed position to its open position. Such movement of the door **70** to its open position causes the closure assembly **60** to transition to the open condition.

(24) The door **70** has an interior side surface **71**, an exterior side surface opposite the interior side surface **71**, a top edge **72**, a bottom edge **73**, a hinge edge **74** at which the door **70** is pivotally mounted to the doorframe, and a free edge **75** opposite the hinge edge. The door **70** also has a door preparation **80** that facilitates the mounting of the exit device **90**. In the illustrated form, the door preparation **80** is formed near the free edge **75**, and includes a channel preparation **81** formed between the interior side surface **71** and the exterior side surface, an upper cavity **82** extending downward from the top edge **72**, and a lower cavity **83** extending upward from the bottom edge **73**. The channel preparation **81** includes an upper channel **84** extending downward from the upper cavity **83**, and a lower channel **85** extending upward from the lower cavity **83**. In certain forms, the upper channel **84** and lower channel **85** may be discrete channels that do not connect to one another. In other forms, the upper channel **84** and lower channel **85** may be provided as upper and lower portions of a contiguous channel preparation **81**.

(25) The door preparation **80** also includes an access arrangement **88** (FIG. 3) that is formed in the interior side surface **71**, and which includes at least one aperture through which the channel preparation **81** is accessible. As a result, each of the cavities **82**, **83** is in communication with the access arrangement **88** via the channel preparation **81**. In the illustrated form, the access arrangement **88** includes an upper aperture **86** and a lower aperture **87**, each of which is in communication with a corresponding one of the cavities **82**, **83** via the channel preparation **81**. More specifically, the upper aperture **86** is in communication with the upper cavity **82** via the upper channel **84**, and the lower aperture **87** is in communication with the lower cavity **83** via the lower channel **85**. In other embodiments, the access arrangement **88** may be provided in the form of a single aperture through which both the upper and lower portions of the channel preparation **81** are accessible.

(26) With additional reference to FIG. 2, the exit device **90** generally includes a pushbar assembly **100**, a remote latching assembly **200** operably connected with the pushbar assembly **100**, and a coordination mechanism **300** that aids in controlling the operation of the remote latching assembly **200**. The pushbar assembly **100** generally includes a mounting assembly **110** configured for mounting to the door **70**, a drive assembly **120** mounted to the mounting assembly **110** for movement between an actuated state and a deactuated state, and a latch control assembly **140** operably connected with the drive assembly **120** via a lost motion connection **108**. As described herein, the drive assembly **120** is biased toward the deactuated state, and is operable to be driven to the actuated state when manually actuated by a user. The latch control assembly **140** also has an actuated state and a deactuated state, and is operably connected with the drive assembly **120** such that actuation of the drive assembly **120** causes a corresponding actuation of the latch control assembly **140**. The pushbar assembly **100** may further include a dogging mechanism **130** operable to selectively retain the drive assembly **120** and the latch control assembly **140** in the actuated states thereof, thereby dogging the exit device **90**.

(27) The remote latching assembly **200** generally includes a connection assembly **210** mounted in the channel preparation **81**, an upper latch mechanism **220** mounted in the upper cavity **82**, and a lower latch mechanism **230** mounted in the lower cavity **83**. As described herein, each of the upper latch mechanism **220** and the lower latch mechanism **230** is operably connected with the pushbar assembly **100** via the connection assembly **210** such that the pushbar assembly **100** is capable of actuating the remote latching assembly **200**.

(28) As described in further detail below, the coordination mechanism **300** aids in coordinating the movement of the latch mechanisms **220**, **230** during operation of the closure assembly **60**. In the

illustrated embodiment, the coordination mechanism **300** is positioned in the pushbar assembly **100**, and includes an anchor bracket **310** securely mounted to the mounting assembly **110**, a cup **320** engaged with the latch control assembly **140**, and a spring **330** that is seated in the cup **320** and engaged with the bracket **310**.

(29) The mounting assembly **110** generally includes an elongated channel member **111**, a base plate **112** mounted in the channel member **111**, and a pair of bell crank mounting brackets **114** coupled to the base plate **112**. The channel member **111** extends along the longitudinal (X) axis **102**, has a width in the lateral (Y) directions, and has a depth in the transverse (Z) directions. Each of the mounting brackets **114** includes a pair of laterally-spaced walls **115** that extend away from the base plate **112** in the forward (Z.sup.+) direction. The illustrated mounting assembly **110** also includes a face plate **113** that encloses a distal end portion of the channel member **111**, a header plate **116** positioned adjacent a proximal end of the channel member **111**, and a header casing **117** mounted to the header plate **116**.

(30) The drive assembly **120** includes a drive rod **122** extending along the longitudinal axis **102**, a pushbar **124** having a pair of pushbar brackets **125** mounted to the rear side thereof, and a pair of bell cranks **126** operably connecting the drive rod **122** with the pushbar **124**. As described herein, the drive rod **122** is mounted for movement in the longitudinal (X) directions, the pushbar **124** is mounted for movement in the transverse (Z) directions, and the bell cranks **126** couple the drive rod **122** and the pushbar **124** for joint movement during actuation and deactuation of the drive assembly **120**. Each bell crank **126** is pivotably mounted to a corresponding one of the bell crank mounting brackets **114**, and includes a first arm that is pivotably connected to the drive rod **122**, and a second arm that is pivotably connected to a corresponding one of the pushbar brackets **125**. The pivotal connections may, for example, be provided by pivot pins **121**. The drive assembly **120** further includes a return spring **127** that is engaged with the mounting assembly **110** and which biases the drive assembly **120** toward its deactuated state.

(31) Each of the drive rod **122** and the pushbar **124** has an actuated position in the actuated state of the drive assembly **120**, and a deactuated position in the deactuated state of the drive assembly **120**. During actuation and deactuation of the drive assembly **120**, the drive rod **122** moves in the longitudinal (X) directions between a proximal deactuated position and a distal actuated position, and the pushbar **124** moves in the transverse (Z) directions between a projected or forward deactuated position and a depressed or rearward actuated position. Thus, during actuation of the drive assembly **120**, the drive rod **122** moves in the distal (X.sup.-) direction, and the pushbar **124** moves in the rearward (Z.sup.-) direction. Conversely, during deactuation of the drive assembly **120**, the drive rod **122** moves in the proximal (X.sup.+) direction, and the pushbar **124** moves in the forward (Z.sup.+) direction. The bell cranks **126** translate longitudinal movement of the drive rod **122** to transverse movement of the pushbar **124**, and translate transverse movement of the pushbar **124** to longitudinal movement of the drive rod **122**. Thus, the longitudinal movement of the drive rod **122** and the transverse movement of the pushbar **124** are bijectively correlated with one another by the bell cranks **126**.

(32) With the drive assembly **120** in its deactuated state, a user may depress the pushbar **124** to transition the drive assembly **120** to its actuated state. As the pushbar **124** is driven toward its depressed position, the bell cranks **126** translate the rearward movement of the pushbar **124** to distal movement of the drive rod **122**, thereby compressing the return spring **127**. When the actuating force is subsequently removed from the pushbar **124**, the spring **127** returns the drive rod **122** to its proximal position, and the bell cranks **126** translate the proximal movement of the drive rod **122** to forward movement of the pushbar **124**, thereby returning the drive assembly **120** to its deactuated state.

(33) The dogging mechanism **130** generally includes a post **132** mounted to the mounting plate **112** and a dogging hook **134** pivotably mounted to the post **132**. When the drive assembly **120** is in its actuated state, an opening formed in the distal end portion of the drive rod **122** is aligned with the

dogging hook **134**. In this state, the dogging hook **134** can be pivoted between a dogging position and a releasing position. In the dogging position, the hook **134** extends into the opening such that the dogging mechanism **130** retains the drive rod **122** in its distal position against the biasing force of the return spring **127**, thereby retaining the drive assembly **120** in its actuated state and dogging the exit device **100**. As the hook **134** is returned to its release position, the hook **134** exits the opening, and the drive rod **122** becomes free to return to its deactuated position under the biasing force of the return spring **127**.

(34) With additional reference to FIG. 3, the latch control assembly **140** generally includes a control link **142** and a yoke **144**, which are coupled for joint movement with one another along the longitudinal (X) axis **102**. The latch control assembly **140** further includes a pair of pivot cranks **146** pivotally mounted to the header plate **116**, and a pair of drivers **150** mounted to the header plate **116** for movement in the lateral (Y) directions. The pair of drivers **150** includes an upper driver **152** and a lower driver **153**, each of which is operably connected with the yoke **144** via a corresponding one of the pivot cranks **146**. Each pivot crank **146** includes a first portion that is pivotably connected to the yoke **144**, and a second portion that is pivotably connected to a corresponding one of the drivers **150**.

(35) The control link **142** is operably connected with the drive assembly **120** via the lost motion connection **108** such that actuation of the drive assembly **120** causes a corresponding actuation of the latch control assembly **140**. Each of the control link **142**, the yoke **144**, the upper driver **152**, and the lower driver **153** has a deactuated position in the deactuated state of the latch control assembly **140**, and an actuated position in the actuated state of the latch control assembly **140**. Each of the control link **142** and the yoke **144** has a proximal deactuated position and a distal actuated position, and moves in the longitudinal (X) directions during actuation and deactuation of the latch control assembly **140**. Each of the drivers **150** has a laterally-outward deactuated position and a laterally-inward actuated position, and moves in the lateral (Y) directions during actuation and deactuation of the latch control assembly **140**.

(36) As used herein, the terms “laterally inward” and “laterally outward” may be used to describe the lateral (Y) directions with reference to the longitudinal (X) axis **102** along which the drive rod **122** and the yoke **144** extend. More specifically, the term “laterally inward” may be used to describe a lateral (Y) direction extending toward the longitudinal (X) axis **102**, and the term “laterally outward” may be used to describe a lateral (Y) direction extending away from the longitudinal (X) axis **102**. Thus, for the upper driver **152**, the laterally inward direction is the downward (Y.sup.-) direction, and the laterally outward direction is the upward (Y.sup.+) direction. For the lower driver **153**, by contrast, the laterally inward direction is the upward (Y.sup.+) direction, and the laterally outward direction is the downward (Y.sup.-) direction.

(37) During actuation and deactuation of the latch control assembly **140**, the pivot cranks **146** convert longitudinal movement of the yoke **144** to lateral movement of the drivers **150** and vice versa. With the latch control assembly **140** in its deactuated state, actuation of the drive assembly **120** causes the control link **142** and the yoke **144** to move in the distal (X.sup.-) direction toward the actuated positions thereof. As the yoke **144** is driven toward its actuated position, the pivot cranks **146** translate the distal movement of the yoke **144** to laterally-inward movement of the drivers **150**, thereby moving the drivers **150** to the actuated positions thereof. With the latch control assembly **140** in its actuated state, the lost motion connection **108** may allow the drive assembly **120** to return to its deactuated state without causing a corresponding deactuation of the latch control assembly **140**. When an appropriate deactuating force is exerted on the latch control assembly **140**, for example by the remote latching assembly **200**, the latch control assembly **140** returns to its deactuated state. During deactuation of the latch control assembly **140**, the yoke **144** and the drivers **150** return to the deactuated positions thereof, and the pivot cranks **146** correlate the laterally-outward movement of the drivers **150** with the proximal movement of the yoke **144** and control link **142**.

(38) Each of the illustrated drivers **150** includes a body portion **154** slidably mounted to the header plate **116** for movement in the lateral (Y) directions, and a lift finger **156** coupled with the body portion **154** via a fastener, such as a screw **158**. The lift finger **156** extends through into the channel preparation **81** via the access arrangement **88** and an opening **155** in the body portion **154**. In the illustrated form, the lift finger **156** of the upper driver **152** extends into the upper channel **84** via the upper aperture **86**, and the lift finger **156** of the lower driver **153** extends into the lower channel **85** via the lower aperture **87**. As described herein, the lift fingers **156** are engaged with the connection assembly **210** such that the latch control assembly **140** and the remote latching assembly **200** are operably connected with one another.

(39) In the illustrated form, the remote latching assembly **200** is of the type often referred to as a “concealed” remote latching assembly. More specifically, the illustrated remote latching assembly **200** is mounted within the door preparation **80** such that the connection assembly **210** and the latching mechanisms **220**, **230** are primarily concealed from view. It is also contemplated that the remote latching assembly **200** may be of the type often referred to as a “surface” remote latching assembly, which is configured for mounting to the interior side surface **71** of the door **70**. In such forms, the door preparation **80** may omit one or more features configured for use with the concealed remote latching assembly **200**, such as the cavities **82**, **83** and/or the channels **84**, **85**.

(40) The connection assembly **210** includes an upper connector **212** and a lower connector **213**, each of which is operably connected with the latch control assembly **140**. Each of the connectors **212**, **213** has a laterally inward portion that is coupled with the lift finger **156** of a corresponding one of the drivers **152**, **153**. More specifically, the upper connector **212** has a lower end **214** coupled with the lift finger **156** of the upper driver **152**, and the lower connector **213** has an upper end **215** coupled with the lift finger **156** of the lower driver **153**. Each of the connectors **212**, **213** also has a laterally outward portion that is coupled with a movable link member of a corresponding one of the latch mechanisms **220**, **230**. More specifically, the upper connector **212** has an upper end **216** coupled to a linkage **226** of the upper latch mechanism **220** (FIG. 4), and the lower connector **213** has a lower end **217** coupled to a linkage **236** of the lower latch mechanism **230** (FIG. 5).

(41) In the illustrated embodiment, the connectors **212**, **213** are provided in the form of flexible cables. More specifically, each of the connectors **212**, **213** is provided as a pull cable that is operable to transmit pulling forces, but which cannot transmit pushing forces. In other embodiments, one or both of the connectors **212**, **213** may be provided in a form that is capable of transmitting pushing forces in addition to pulling forces. Such push/pull forms of connectors may be provided in a form that is flexible, such as a sheathed or Bowden cable, or may alternatively be provided in a form that is rigid, such as a rigid rod.

(42) With additional reference to FIG. 4, the upper latch mechanism **220** generally includes a housing **222**, a latchbolt **224** mounted to the housing **222** for pivotal movement between a latching position and an unlatching position, the linkage **226** to which the upper end **216** of the upper connector **212** is coupled, and a blocking member **227** mounted to the housing **222** for pivotal movement between a blocking position and an unblocking position. The linkage **226** is pivotably connected to the blocking member **227** such that lateral movement of the linkage **226** is correlated with the pivotal movement of the blocking member **227**. More specifically, downward or laterally inward movement of the linkage **226** is correlated with movement of the blocking member **227** toward its unblocking position, and upward or laterally outward movement of the linkage **226** is correlated with movement of the blocking member **227** toward its blocking position. The upper latch mechanism **220** also includes a spring **228** urging the linkage **226** in the upward or laterally outward direction, thereby biasing the blocking member **227** toward its blocking position.

(43) When in its blocking position, the blocking member **227** retains the latchbolt **224** in its latching position. As the upper connector **212** pulls the linkage **226** in the downward or laterally inward direction against the force of the spring **228**, the blocking member **227** moves toward its unblocking position, and the latchbolt **224** becomes free to move to its unlatching position. When

in its unlatching position, the latchbolt **224** retains the blocking member **227** in its blocking position against the biasing force of the spring **228**. More specifically, a retaining member **229** formed on the latchbolt **224** projects into the path along which the blocking member **227** travels when moving from its unblocking position to its blocking position, thereby preventing such travel of the blocking member **227**. When the latchbolt **224** returns to its latching position, the biasing member **228** returns the blocking member **227** to its blocking position, and the latchbolt **224** is once again retained in its latching position.

(44) Also illustrated in FIG. **4** is an upper strike **202** configured to be mounted to the upper jamb of the doorframe in which the door **70** is installed. In certain embodiments, the upper strike **202** may be considered to constitute a portion of the frame. The upper strike **202** includes a projection **203** which, when the door **70** is in its closed position, extends into a channel **225** formed in the latchbolt **224**. When the door **70** is pushed in its opening direction, the projection **203** urges the latchbolt **224** toward its unlatching position. When the blocking member **227** is in its blocking position, such unlatching movement of the latchbolt **224** is prevented, and engagement between the latchbolt **224** and the upper strike **202** prevents opening movement of the door **70**. When the blocking member **227** is in its unblocking position, opening movement of the door **70** drives the latchbolt **224** toward its unlatching position.

(45) With the door **70** in its open position, the latchbolt **224** is maintained in its unlatching position, for example due to a biasing member or gravity urging the latchbolt **224** toward its unlatching position. Thus, the latchbolt **224** retains the blocking member **227** in its unblocking position while the door **70** is open. When the door **70** is returned to its closed position, the projection **203** enters the channel **225** and returns the latchbolt **224** to its latching position. As the latchbolt **224** returns to its latching position, the retaining member **229** pivots therewith, thereby freeing the blocking member **227** to travel toward its blocking position. The spring **228** thus returns the blocking member **227** to its blocking position, thereby driving the upper end **216** of the upper connector **212** in the upward or laterally-outward direction.

(46) With additional reference to FIG. **5**, the lower latch mechanism **230** generally includes a housing **232**, a deadbolt **234** mounted to the housing **232** for movement between an extended position and a retracted position, a linkage or traveler **236** movably mounted to the housing **232**, and a biasing member **238** urging the traveler **236** in the downward or laterally-outward direction. Also illustrated in FIG. **5** is a lower strike **204** configured to be mounted to the lower jamb or floor of the doorframe in which the door **70** is installed. In certain embodiments, the lower strike **204** may be considered to constitute a portion of the frame. The strike **204** includes a pocket **205** which, when the door **70** is in its closed position, is operable to receive the lower end of the deadbolt **234** when the deadbolt **234** is in its extended position.

(47) The traveler **236** is engaged with the deadbolt **234** such that an externally-applied pushing force exerted on the bottom of the deadbolt **234** actuates a deadlocking mechanism **239** of the lower latch mechanism **230**. More specifically, such a pushing force on the deadbolt **234** drives the traveler **236** into engagement with the housing **232**, thereby preventing further laterally-inward movement of the deadbolt **234**. The traveler **236** is coupled to the lower end **217** of the lower connector **213** such that the traveler **236** retracts the deadbolt **234** in response to movement of the lower connector **213** in the upward or laterally-inward direction. Thus, retraction or laterally-inward movement of the lower connector **213** causes the deadbolt **234** to exit the pocket **205**, thereby permitting opening of the door **70**. As described herein, the lower connector **213** is retained in this retracted position until the door **70** returns to its closed position. When the lower connector **213** subsequently becomes free to move in the laterally-outward direction, the biasing member **238** drives the traveler **236** downward, thereby causing a corresponding downward or laterally-outward movement of the lower end **217** of the lower connector **213**. Such downward movement of the traveler **236** also drives the deadbolt **234** to its extended position, thereby causing the end portion of the deadbolt **234** to enter the pocket **205**.

(48) With additional reference to FIGS. 6 and 7, the coordination mechanism **300** is configured to selectively retain the latch control assembly **140** in its actuated state, and generally includes an anchor bracket **310**, a cup **320**, and a spring **330**. In the illustrated embodiment, the coordination mechanism **300** is mounted in the pushbar assembly **100** in the vicinity of the lost motion connection **108**, and exerts a proximal biasing force on the control link **142**. Additionally, the illustrated control link **142** is provided in the form of a split link or a fork link, and includes two longitudinally-extending arms **143** that are offset from one another in the lateral (Y) directions, a base plate **147** connecting the arms **143** to one another, and a shoulder **148** extending from the base plate **147**. The arms **143**, base plate **147**, and shoulder **148** cooperate to define a receiving space **149** that is sized and shaped to receive the cup **320**.

(49) The anchor bracket **310** is securely mounted to the mounting assembly **110**, and in certain embodiments, may be considered to be included in the mounting assembly **110**. The illustrated bracket **310** includes a base wall **312** and a pair of sidewalls **314** extending from opposite ends of the base wall **312** such that a receiving space **319** is defined therebetween. The proximal end portion of the bracket **310** also includes a flange **316** that extends from the base wall **312** such that a pair of channels **315** are formed between the flange **316** and the sidewalls **314**. Thus, the proximal end portion of the bracket **310** has a generally E-shaped cross-section in which the receiving space **319** is divided into two channels **315**, and the distal end portion of the bracket **310** has a generally C-shaped cross-section in which the receiving space **319** defines a single channel **313**.

(50) The cup **320** is sized and shaped to be received in the receiving spaces **149**, **319**, and defines a chamber **321** sized and shaped to receive the spring **330**. The cup **320** includes an end wall **322** and a sleeve **324**, which cooperate to at least partially define the chamber **321**. The sleeve **324** defines at least one slot **326**, through which the flange **316** extends into the chamber **321**. In the illustrated form, a pair of diametrically opposite slots **326** extend distally from the proximal end of the sleeve **324**, and the flange **316** is partially received in each of the slots **326**.

(51) With the coordination mechanism **300** installed to the pushbar assembly **100**, the bracket **310** is secured to the mounting plate **112** such that the control link **142** extends therethrough. By way of example, the mounting plate **112** may include a series of apertures **107**, and spring clips or snap-fit posts **317** may extend from the ends of the sidewalls **314** and matingly engage two or more of the apertures **107**. Each of the arms **143** extends through a corresponding one of the channels **315** such that the flange **316** extends into the gap between the arms **143**, thereby supporting and guiding the cup **320** as the cup **320** moves longitudinally.

(52) At least a portion of the base plate **147** is positioned between the sidewalls **314** such that the receiving spaces **149**, **319** intersect one another, and the cup **320** is seated in the receiving spaces **149**, **319** and captured between the control link **142** and the bracket **310**. More specifically, the cup **320** is transversely captured between the base plate **147** and the base wall **312**, and is laterally captured between the arms **143**. Additionally, the end wall **322** is longitudinally captured between the shoulder **148** and the flange **316**, which extends into the chamber **321** via one of the slots **326**. The spring **330** is seated in the chamber **321** and captured between the flange **316** and the end wall **322**, thereby distally urging the end wall **322** into contact with the shoulder **148** and resisting movement of the control link **142** in the proximal (X.sup.+) direction. Thus, the coordination mechanism **300** is operable urge the latch control assembly **140** toward its actuated state, and to resist movement of the latch control assembly **140** toward its deactuated state.

(53) With additional reference to FIG. 8, illustrated therein are certain forces that may be exerted on the latch control assembly **140** during operation of the exit device **90**. More specifically, FIG. 8 illustrates a set of actuating forces **180** urging the latch control assembly **140** toward its actuated state, and a set of deactuating forces **190** urging the latch control assembly **140** toward its deactuated state. As will be appreciated, when the actuating forces **180** are overcome by the deactuating forces **190**, the latch control assembly **140** will be driven toward its deactuated state.

Conversely, when the actuating forces **180** overcome the deactuating forces **190**, the latch control assembly **140** will be driven toward its actuated state.

(54) The actuating forces **180** include an upper driver actuating force **182** acting on the upper driver **152**, a lower driver actuating force **183** acting on the lower driver **153**, and a control link actuating force **186** acting on the control link **142**. Each of the driver actuating forces **182**, **183** urges the corresponding driver **152/153** in its laterally inward actuating direction. More specifically, the upper driver actuating force **182** urges the upper driver **152** in the downward (Y-) direction, and the lower driver actuating force **183** urges the lower driver **153** in the upward (Y+) direction. The control link actuating force **186** urges the control link **142** in its distal (X-) actuating direction. By way of non-limiting example, factors that may contribute to the control link actuating force **186** include actuation of the drive assembly **120**, and the biasing force of the spring **330**.

(55) As noted above, the moving components of the latch control assembly **140** are operably connected with one another substantially without lost motion such that the movements and positions of the components are correlated with one another. As a result, an externally-applied input actuating force **188** exerted on one component of the latch control assembly **140** causes a corresponding resultant actuating force **189** to be exerted on each of the other components. In the illustrated form, the input actuating force **188** is typically provided as the control link actuating force **186**, and may be exerted by one or more components external to the latch control assembly **140**, such as the drive assembly **120** and/or the coordination mechanism **300**. Such an input actuating force **186**, **188** is transmitted to the drivers **150** by the yoke **144** and pivot cranks **146**, thereby providing resultant forces **189** in the form of the driver actuating forces **182**, **183**.

(56) The deactuating forces **190** include an upper driver deactuating force **192** acting on the upper driver **152**, a lower driver deactuating force **193** acting on the lower driver **153**, and a control link deactuating force **196** urging the control link **142** in its proximal (X+) deactuating direction. Each of the driver deactuating forces **192**, **193** urges the corresponding driver **152/153** in its laterally outward deactuating direction. More specifically, the upper driver deactuating force **192** urges the upper driver **152** in the upward (Y+) direction, and the lower driver deactuating force **193** urges the lower driver **153** in the downward (Y-) direction. By way of non-limiting example, factors that may contribute to the driver deactuating forces **192**, **193** include the biasing forces of the springs **228**, **238** of the latch mechanisms **220**, **230**.

(57) As with the actuating forces **180**, an externally-applied input deactuating force **198** exerted on one component of the latch control assembly **140** causes a corresponding resultant deactuating force **199** to be exerted on each of the other components. In the illustrated form, the input deactuating force **198** is typically provided as the upper driver deactuating force **192** and/or the lower driver deactuating force **193**, each of which may be exerted by the corresponding spring **228/238** via the corresponding connector **112/113**. For example, providing the input deactuating force **198** as the lower driver deactuating force **193** causes a resultant deactuating force **199** in the form of the control link deactuating force **196**. When both the upper driver **152** and lower driver **153** are being pulled laterally outward by the remote latching assembly **200**, both of the driver deactuating forces **192**, **193** contribute to input deactuating force **198**, thereby exerting an increased resultant force **196**, **199** on the control link **142**.

(58) With additional reference to FIG. **9**, illustrated therein are the coordination mechanism **300** and a portion of the pushbar assembly **100** in various states corresponding to the above-described conditions of the closure assembly **60**. FIG. **9a** corresponds to the secured condition, in which the drive assembly **120** and the latch control assembly **140** are in the deactuated states thereof, and the coordination mechanism **300** is in a compressed state. FIG. **9b** corresponds to the unsecured condition, in which the drive assembly **120** and the latch control assembly **140** are in the actuated states thereof, and the coordination mechanism **300** is in an expanded state. FIG. **9c** corresponds to the open condition, in which the drive assembly **120** is in its deactuated state, the latch control assembly **140** is in its actuated state, and the coordination mechanism is **300** in its expanded state.

(59) FIG. 9a illustrates the coordination mechanism 300 with the closure assembly 60 in the secured condition, in which the exit device 90 is deactuated, and the remote latching assembly 200 engages the frame and retains the door 70 in its closed position. In this state, each of the latch mechanisms 220, 230 is in its deactuated state and is engaged with the corresponding one of the strikes 202, 204. With the upper latch mechanism 220 in its deactuated state, the blocking member 227 retains the latchbolt 224 in its latching position, thereby maintaining engagement between the latchbolt 224 and the projection 203. Additionally, the deadlocking features of the lower latch mechanism 230 prevent external pushing forces from driving the deadbolt 234 to its retracted position, thereby preventing the tip of the deadbolt 234 from exiting the pocket 205.

(60) With the closure assembly 60 in its secured condition, the latch control assembly 140 is in its deactuated state. With the control link 142 in its proximal deactuated position, the spring 330 is compressed between the shoulder 148 and the flange 316. As a result, the coordination mechanism 300 exerts an actuating force 186 on the control link 142, thereby urging the latch control assembly 140 toward its actuated state. Additionally, the springs 228, 238 of the latch mechanisms 220, 230 place the connectors 212, 213 in tension and exert driver deactuating forces 192, 193 that pull the drivers 150 laterally outward. Thus, both the upper latch mechanism spring 228 and the lower latch mechanism spring 238 contribute to the deactuating input force 198, which results in a resultant control link deactuating force 196, 199 that exceeds the actuating control link force 186 exerted by the coordination mechanism 300. As a result, the control link 142 is retained in its proximal position, and the latch control assembly 140 remains in its deactuated state.

(61) From the secured condition, the closure assembly 60 can be transitioned to the unsecured condition by driving the pushbar 124 to its depressed position, thereby actuating the drive assembly 120 and causing a corresponding actuation of the latch control assembly 140. As the latch control assembly 140 moves toward its actuated state, the control link 142 moves toward its distal actuated position, thereby permitting expansion of the spring 330. Additionally, the drivers 150 pull the connectors 212, 213 in the laterally inward actuating direction, and the connectors 212, 213 pull the link members 226, 236 laterally inward against the laterally outward biasing forces of the springs 228, 238, thereby actuating the latch mechanisms 220, 230. With the latch mechanisms 220, 230 in the actuated states thereof, the blocking member 227 is in its unblocking position, and the deadbolt 234 is in its retracted position. As a result, the remote latching assembly 200 does not prevent opening movement of the door 70, and the closure assembly 60 is in the unsecured condition.

(62) With the closure assembly 60 in the unsecured condition, the door 70 is in its closed position, the exit device 90 is actuated, and the coordination mechanism 300 is in the state illustrated in FIG. 9b. From the unsecured condition, the door 70 is free to move toward its open position, and the closure assembly 60 can be transitioned to either of the secured condition and the open condition. In order to transition the closure assembly 60 to the secured condition, the user may release the pushbar 124, thereby causing the return spring 127 to return the drive assembly 120 to its deactuated state. Due to the fact that the drive assembly 120 is connected to the latch control assembly 140 via the lost motion connection 108, such deactuation of the drive assembly 120 does not necessarily cause a corresponding deactuation of the latch control assembly 140.

(63) As indicated above, the lost motion connection 108 prevents the drive rod 122 from pushing the control link 142. Accordingly, deactuation of the drive assembly 120 does not exert the deactuating input force 198 required to return the latch control assembly 140 to its deactuated state against the actuating input force 188 exerted by the coordination mechanism 300. With the door 70 in the closed position, however, the deactuating input force 198 is provided in the form of driver deactuating forces 192, 193 exerted by the remote latching assembly 200. More specifically, the upper driver deactuating force 192 is exerted by the upper connector 112 as the spring 228 drives the linkage 226 to return the blocking member 227 to its blocking position, and the lower driver deactuating force 193 is exerted by the lower connector 113 as the spring 238 drives the traveler 236 to return the deadbolt 234 to its extended position. The resultant control link deactuating force

196, 199 is sufficient to overcome the control link actuating force **186** exerted by the coordination mechanism **300**. As a result, the latch control assembly **140** returns to its deactuated state.

(64) The closure assembly **60** can also be transitioned from the unsecured condition to the open condition. In order to effect such a transition, the user may urge the door **70** toward its open position, for example by exerting a pushing force on the depressed pushbar **124**. As the door **70** moves toward its open position, the projection **203** of the upper strike **202** drives the latchbolt **224** toward its unlatching position. With the door **70** in its open position and the latchbolt **224** in its unlatching position, the user may release the pushbar **124**, thereby causing the drive assembly **120** to return to its deactuated state under the force of the return spring **127**. As the drive rod **122** moves from its distal actuated position (FIG. **9b**) to its proximal deactuated position (FIG. **9c**), the lost motion connection **108** permits the control link **142** to remain in its distal actuated position.

(65) With the closure assembly **60** in the open condition, the spring **238** of the lower latch mechanism **230** urges the traveler **236** in its laterally outward or downward direction. As a result, the lower connector **213** exerts a deactuating input force **193, 198** on the lower driver **153**, which causes a resultant deactuating force **192, 199** urging the upper driver **152** in its laterally outward or upward direction. Additionally, the latchbolt **224** is in its unlatching position and retains the blocking member **227** in its blocking position, thereby preventing the spring **228** from driving the linkage **226** in its laterally outward or upward direction. As a result, the input deactuating force **192, 198** exerted on the upper driver **152** is reduced or eliminated.

(66) As will be appreciated, should the lower driver **153** be permitted to move to its deactuated position, the spring **238** will drive the deadbolt **234** to its extended position. If the door **70** is open when this occurs, the deadbolt **234** may strike the floor, and may drag along the floor as the door **70** moves. Both striking and dragging are typically considered undesirable, and can result in objectionable noise and damage to the latch mechanism **230** and/or the floor. However, these results may be obviated by the coordination mechanism **300**, which retains the latch control assembly **140** in its deactuated state while the closure assembly **60** is in the open condition.

(67) As noted above, when the closure assembly **60** is in the open condition, the lower connector **213** pulls the lower driver **153** in its laterally outward (downward) direction under the force of the lower latch mechanism spring **238**, but the upper connector **212** does not pull the upper driver **152** in its laterally outward (upward) direction. The deactuating input force **198** therefore includes the lower driver deactuating force **193**, but does not include the upper driver deactuating force **192**. As such, the resultant control link deactuating force **196, 199** is lower in the open condition than in the unsecured condition, in which both the upper latch mechanism spring **228** and the lower latch mechanism spring **238** contribute to the deactuating input force **198**. Additionally, the characteristics (e.g., size, shape, and stiffness) of the lower latch mechanism spring **238** and the coordination mechanism spring **330** are selected such that when the coordination mechanism **300** is in the expanded state illustrated in FIG. **9c**, the actuating input force **188** exerted by the spring **330** exceeds the deactuating control link force **196** imparted by the lower latch mechanism spring **238**. The coordination mechanism **300** thus retains the latch control assembly **140** in its actuated state, thereby retaining the lower driver **153** in its actuated state and preventing deactuation of the lower latch mechanism **230**.

(68) From the open condition (FIG. **9c**), the closure assembly **60** can be returned to the secured condition (FIG. **9a**) by closing the door **70**. As the door **70** approaches its closed position, the projection **203** of the upper strike **202** returns the latchbolt **224** to its latching position, thereby freeing the blocking member **226** to return to its blocking position. As the spring **228** drives the linkage **226** laterally outward toward its deactuated position, the upper connector **212** is placed in tension and pulls the upper driver **152** upward, thereby exerting an upper driver deactuating force **192**. Thus, both the upper latch mechanism spring **228** and the lower latch mechanism spring **238** contribute to the deactuating input force **198**, thereby increasing the resultant deactuating force **196, 199** on the control link **142**.

(69) The characteristics (e.g., size, shape, and stiffness) of the latch mechanism springs **228**, **238** and the coordination mechanism spring **330** are selected such that when both the upper latch mechanism spring **228** and the lower latch mechanism spring **238** contribute to the deactuating input force **198**, the resultant control link deactuating force **196**, **199** is sufficient to drive the coordination mechanism **300** from its expanded state (FIG. **9c**) to its compressed state (FIG. **9a**). With the deactuating forces **190** imparted by the remote latching assembly **200** exceeding the actuating forces **180** imparted by the coordination mechanism **300**, the latch control assembly **140** and the remote latching assembly **200** return to the deactuated states thereof, and the closure assembly **60** transitions to the secured condition.

(70) Due to the fact that the bracket **310** is secured to the mounting assembly **110**, the proximal end of the spring **330** is provided with an anchor point having a fixed location within the exit device **90**. The compression displacement of the spring **330** (and thus the force exerted on the control link **142**) is therefore correlated with the position of the control link **142**, and is independent of the actuated/deactuated state of the drive assembly **120**. As a result, the coordination mechanism **300** maintains a consistent functionality regardless of whether or not the pushbar **124** is in its depressed position.

(71) In the illustrated embodiment, the coordination mechanism **300** biases the latch control assembly **140** toward its actuated state by exerting an input actuating force **186**, **188** on the control link **142**. In other embodiments, the coordination mechanism **300** may exert the input actuating force **188** another component of the latch control assembly **140**. By way of illustration, the exit device **90** may include one or more header-mounted coordination mechanisms in addition or as an alternative to the illustrated coordination mechanism **300**. In such a header-mounted version of the coordination mechanism **300**, the anchor bracket may be secured to the header plate **117**, and the spring **330** may be engaged with one of the upper driver **152** or the lower driver **153**. Additionally or alternatively, the compression spring **330** may be replaced with a torsion spring, which may be engaged between the header plate **117** and a pivot crank **146**. Where multiple coordination mechanisms **300** are used, lighter springs may need to be selected to ensure that the total input actuating force **188** remains within the range required to selectively prevent deactuation of the latch control assembly **140** in the manner described above.

(72) As is evident from the foregoing, the coordination mechanism **300** aids in coordinating the movement of the latch mechanisms **220**, **230** according to a desired sequence of events. More specifically, the coordination mechanism **300** aids in preventing the lower latch mechanism **230** from moving to its deactuated state while the upper latch mechanism **220** is in its deactuated state, and in causing the lower latch mechanism **230** to move to its deactuated state in response to deactuation of the upper latch mechanism **220**. As noted above, the upper latch mechanism **220** will typically return to its deactuated state only when the door **70** returns to its closed position, in which the lower latch mechanism **230** is aligned with the pocket **205**. Thus, the coordination mechanism **300** aids in preventing premature extension of the deadbolt **234** and the undesirable consequences thereof, such as dragging along and/or striking the floor.

(73) Additionally, the illustrated coordination mechanism **300** is capable of providing the above-described coordination of bolt movement without requiring the transmission of pushing forces between the pushbar assembly **100** and the remote latch mechanisms **220**, **230**, and of doing so while installed within the pushbar assembly **100**. These capabilities may provide for certain advantages, such as facilitating customization of the exit device **90**. For example, while the exit device **90** is illustrated as including a concealed remote latching assembly **200** in which the connectors **212**, **213** are provided as pull cables, a pushbar assembly **100** including the coordination mechanism **300** may alternatively be used in combination with a remote latching assembly **200** that is surface-mounted and/or includes push/pull connectors, such as Bowden cables or rigid rods. Thus, a single pushbar assembly **100** may be suitable for use with several different configurations of remote latching assemblies without requiring significant modification, which may reduce the

costs associated with the manufacture, inventory, and/or installation of exit devices of varying formats.

(74) The above-mentioned features may also aid in providing the door **70** with increased structural integrity. For example, certain existing exit devices including concealed remote latching assemblies involve the use of rigid rods, which require relatively large channels within the door. In obviating the need for the transmission of pushing forces, however, the coordination mechanism **300** may facilitate the use of flexible pull cables that can be used with a smaller channel preparation, thereby increasing the amount of material that can be provided on either side of the channels. While other existing exit devices include concealed remote latching assemblies having flexible pull cables, some such exit devices coordinate the movement of the latch mechanisms using a device that is installed within a mortise-like cutout in the door. The need for such a mortise-like cutout may be obviated by the coordination mechanism **300**, for example when the coordination mechanism **300** is provided within the pushbar assembly **100**. With the need for such a cutout obviated, the free edge **75** may be substantially unbroken by the door preparation, which may further increase the structural integrity of the door **70**.

(75) With reference to FIG. **10**, illustrated therein is a header-mounted coordination mechanism **400** according to certain embodiments, along with a coupling assembly **500** according to certain embodiments. As described herein, the coordination mechanism **400** functions in a manner similar to that described above with reference to the coordination mechanism **300**, and the coupling assembly **400** facilitates the coupling of the latch control assembly **140** and the remote latching assembly **200**.

(76) In certain embodiments, the header-mounted coordination mechanism **400** maybe used in combination with the above-described coordination mechanism **300**, while in other embodiments the coordination mechanism **400** may be used independently of the coordination mechanism **300**. The coordination mechanism **400** generally includes an anchor bracket **410** secured to the mounting assembly **110**, a cup bracket **420** secured to the yoke **144**, and one or more springs **430** engaged between the anchor bracket **410** and the cup bracket **420**. While the illustrated embodiment includes three springs **430**, it is to be appreciated that more or fewer springs may be utilized in other embodiments.

(77) The anchor bracket **410** includes a plurality of bosses **412**, and the cup bracket **420** includes a plurality of cups **422** aligned with the bosses **412**. Additionally, a proximal end portion of each spring **430** is engaged with a boss **412**, and a distal end portion of each spring **430** is seated in a cup **422**. The springs **430** are thus captured between the anchor bracket **410** and the cup bracket **420**, and are operable to exert a distal biasing force on the yoke **144**. In this manner, the coordination mechanism **400** is operable to bias the latch control assembly **140** toward its actuated state and to resist movement of the latch control assembly **140** toward its deactuated state. The coordination mechanism **400** thereby operates in a manner analogous to that described above with reference to the coordination mechanism **300** such that the coordination mechanism **400** is operable to selectively retain the lower latch mechanism **230** in its actuated state, and to allow the lower latch mechanism **230** to return to its deactuated state in response to deactuation of the upper latch mechanism **220**.

(78) With additional reference to FIGS. **11** and **12**, the coupling assembly **500** is configured to be used in place of the above-described lift finger **156**, and may provide for increased ease of installation relative to the arrangement illustrated in FIG. **3**. Also illustrated in FIGS. **11** and **12** is a cable **502** that can be connected to one of the latch mechanisms **220**, **230** to serve the functions of the corresponding connector cable **212**, **213**. The coupling assembly **500** generally includes a guide member **510** configured for mounting to the driver **150**, a retaining member **520** that aids in retaining the guide member **510** relative to the driver **150**, and a compression fitting **530** seated in the guide member **510**.

(79) The guide member **510** generally includes a body portion **512** having a cavity **513** and an

opening **514**, and a cable guide **516** having a through-passage **517** connected with the cavity **513**. The retaining member **520** includes a body portion **522** having a flat **523** formed thereon, a post **524** formed on one end of the body portion **522**, and an arm **526** formed on the opposite end of the body portion **522**. The guide member **510** is seated in the opening **155** of the driver **150** such that the cable guide **516** extends beyond the rear surface of the header plate **116**. The body portion **522** of the retaining member **520** extends through another opening in the driver **150** such that a portion of the driver **150** is captured between the arm **526** and the guide member body portion **512**. Additionally, the post **524** is received in the opening **514** such that the retaining member **520** restrains the guide member **510** from moving transversely. This arrangement is maintained by a screw **504**, which is threaded through the captured portion of the driver **150** and is engaged with the flat **523**.

(80) The compression fitting **530** includes a first member **532** that is seated in the cavity **513**, a second member **534** rotatably mounted to the first member **532**, a manually-graspable turn piece **535** formed on the second member **534**, and a clamp **536** mounted in the first member **532** and engaged with the second member **534**. The cable **502** passes through the compression fitting **530** such that a portion of the cable **502** is received in the clamp **536**, and such that the cable **502** exits the compression fitting **530** via an opening in the turnpiece **535**. The internal surfaces of the clamp **536** include ridges that grip the cable **502** when the clamp **536** is compressed, and which release the cable **502** when the clamp **536** is expanded. The clamp **536** is engaged with the second member **534** such that rotation of the second member **534** in opposite directions compresses and expands the clamp **536**.

(81) With the exit device **100** mounted to the door **70**, the cable guide **516** extends through the door preparation **80** and directs the cable **502** toward the appropriate one of the latch mechanisms **220**, **230**. Thus, like the lift finger **156**, the coupling assembly **500** facilitates the coupling of the remote latching assembly **200** with the latch control assembly **140**. Unlike the arrangement illustrated in FIG. **3**, however, the coupling assembly **500** also directs the cable **502** to extend through the door preparation **80** and the latch control assembly **140** to a location at which the end of the cable **502** is more readily accessible, thereby facilitating the installation process. As described herein, the installation process is further facilitated by other features of the coupling assembly **500**, such as the compression fitting **530**.

(82) With additional reference to FIG. **13**, illustrated therein is an example process **600** for installing the exit device **100** to the door **70** using the coupling assembly **500**. Operations illustrated for the processes in the present application are understood to be examples only, and operations may be combined or divided, and added or removed, as well as re-ordered in whole or in part, unless explicitly stated to the contrary.

(83) The process **600** may begin with an operation **602** that generally involves partially installing the remote latching assembly **200** to the door **70**. More particularly, the operation **602** involves installing the upper latch mechanism **220** to the upper cavity **82** and running the upper connector **212** through the upper channel **84** and out the upper aperture **86**. Similarly, the operation also involves installing the lower latch mechanism **230** to the lower cavity **83** and running the lower connector **213** through the lower channel **85** and out the lower aperture **87**.

(84) The process **600** also includes an operation **604** that generally involves installing two instances of the coupling assembly **500** to the latch control assembly **140**. One of the coupling assemblies **500** is mounted to one of the drivers by inserting the guide member **510** through the opening **155** in the driver **150** such that the cable guide **516** extends beyond the rear surface of the header plate **116**. The compression fitting **530** is coupled to the guide member **510** such that the turnpiece **535** is accessible from the exposed front side of the latch control assembly **140**. The retaining member **520** is then inserted and the screw **504** is installed to affix the coupling assembly **500** to the driver **150** in the manner described above. These steps are then repeated to install and affix the other coupling assembly **500** to the other driver **150**.

(85) After installing the remote latching assembly **200** to the door **80** and installing the coupling assemblies **500** to the latch control assembly **140**, the process **600** includes an operation **606**, which generally involves passing the cables **212**, **213** through the coupling assemblies **600**, as illustrated in FIG. **14**. This operation **606** may begin with positioning the device close to the door **70**, and inserting an end portion **503** of each cable **502** into the opening **517** of the appropriate cable guide **516**. As the user urges the cables **502** through the coupling mechanisms **500**, the end portions **503** emerge from the exposed ends of the compression fittings **530**, which are in a loosened state so as to permit the cables **502** to pass through the clamps **536**. The device is then placed against the door **70** such that the cable guides **516** extend through the apertures **86**, **87**, and the end portions **503** are pulled to remove slack from the cables **502**.

(86) Following the operation **606**, the process **600** proceeds to an operation **608**, which generally involves mounting the pushbar assembly **100** to the door **70**. The process **600** also includes an operation **610**, which generally involves dogging the exit device **100**. More particularly, the operation **610** involves depressing the pushbar **124** to actuate the drive assembly **120**, and manipulating the dogging mechanism **130** to retain the drive assembly **120** in its actuated state. With the pushbar assembly **100** mounted to the door **70** and in the dogged state, the process **600** may proceed to operations **612**, **614**, which generally involve securing the cables **502** to the drivers **150** when the latch mechanisms **220**, **230** are in predetermined states.

(87) The operation **612** begins with the upper latch mechanism **220** in its deactuated state, in which the latchbolt **224** is in its latching position, and the blocking member **227** is in its blocking position. In this state, the user urges the latchbolt **224** toward its unlatching position while slowly pulling the exposed portion of the upper cable **212**. The blocking member **227** retains the latchbolt **224** in its latching position until the blocking member **227** reaches its unblocking position, at which point the latchbolt **224** moves to its unlatching position under the force applied by the user. This movement of the latchbolt indicates to the user that the upper latching mechanism **220** has reached the actuated state corresponding to the actuated state of the drive assembly **120**, and that the cable **212** is of the appropriate effective length. The user then operates the compression fitting **530** to secure the cable **212** to the driver **152** while the cable **212** is of the appropriate effective length. More specifically, the user rotates the second member **534** using the turnpiece **535** to compress the clamp **536** such that the clamp **536** is secured to the cable **212**. In this state, the upper coupling assembly **500** secures the cable **212** to the upper driver **152** and retains the appropriate effective length of the cable **212**.

(88) The operation **614** is similar to the operation **612**, and begins with the lower latch mechanism **230** in the deactuated state, in which the deadbolt **234** is in its extended position. The user slowly pulls the exposed portion of the lower cable **213** until the deadbolt **234** is capable of clearing the pocket **205** and will not drag on the floor. This position indicates to the user that the lower latching mechanism **230** has reached the actuated state corresponding to the actuated state of the drive assembly **120**, and that the cable **213** is of the appropriate effective length. The user then operates the compression fitting **530** to secure the cable **213** to the driver **153** in a manner similar to that described above with reference to the operation **612**. With the operation **614** completed, the lower coupling assembly **500** secures the cable **213** to the lower driver **153** and retains the appropriate effective length of the cable **213**.

(89) After completing the operations **612**, **614**, the process **600** proceeds to an operation **616**, which generally involves trimming off the exposed end portions **503** of the cables **212**, **213**. The operation **616** may involve leaving a certain excess length to facilitate later adjustment of the effective lengths of the cables **212**, **213**. Such adjustment may also be facilitated by the compression fittings **530**, which can be loosened to release the corresponding cable **502** and subsequently tightened when the appropriate effective length has been obtained.

(90) With reference to FIG. **15**, illustrated therein is a coordination mechanism **700** according to certain embodiments. The coordination mechanism **700** includes a first bracket **710**, a second

bracket **720** slidably engaged with the first bracket **710**, and a spring **730** engaged between the first and second brackets **710**, **720**. The first bracket **710** defines a chamber **712** that is delimited by an end wall **713**, and which has a slot **714** formed in one side thereof. The second bracket **720** includes an arm **722** that projects into the chamber **712** via the slot **722**, and which has a post **723** projecting therefrom. The spring **730** is seated in the chamber **720** and is engaged between the brackets **710**, **720**. More particularly, the lower end **731** of the spring **730** is engaged with the end wall **713**, and the upper end portion **732** of the spring **730** is mounted to the post **723**.

(91) With additional reference to FIG. **16**, illustrated therein is the coordination mechanism **700** installed to the pushbar assembly **100**, portions of which have been omitted for clarity. The first bracket **710** is secured to the mounting assembly **110** and the second bracket **720** is secured to the lower driver **152** such that the spring **730** biases the second bracket **720** in the upward direction. As a result, the coordination mechanism **700** urges the lower driver **152** in its upward actuating direction and resists movement of the lower driver **152** in its downward deactuating direction. The coordination mechanism **700** is therefore operable to function in a manner substantially similar to the above-described coordination mechanisms **300**, **400**, with the difference that the component on which the coordination mechanism **700** acts is the lower driver **152**. This arrangement may be capable of taking up a greater degree of slack or play between the upper latch mechanism and the driver **152**, which may be advantageous in certain circumstances.

(92) Also illustrated in FIG. **16** is a connection assembly **750** according to certain embodiments. The connection assembly **750** includes a lift finger **752** that is coupled to the driver **152**, and which extends through the header plate **116** and terminates in a pair of jaws **753**. The jaws **753** are clamped onto a carriage **754** to which the upper end **215** of the lower cable **213** is coupled. An adjustment mechanism **756** facilitates adjustment of the vertical position of the lift finger **752** and carriage **754**. More particularly, a screw **757** can be loosened to allow for adjustment of the vertical position of the lift finger **752**, for example to remove slack from the cable **213**, and can be tightened to secure the lift finger **752** in a selected location.

(93) While the invention has been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only the preferred embodiments have been shown and described and that all changes and modifications that come within the spirit of the inventions are desired to be protected. It should be understood that while the use of words such as preferable, preferably, preferred or more preferred utilized in the description above indicate that the feature so described may be more desirable, it nonetheless may not be necessary and embodiments lacking the same may be contemplated as within the scope of the invention, the scope being defined by the claims that follow. In reading the claims, it is intended that when words such as “a,” “an,” “at least one,” or “at least one portion” are used there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. When the language “at least a portion” and/or “a portion” is used the item can include a portion and/or the entire item unless specifically stated to the contrary.

Claims

1. A pushbar assembly, comprising: a drive assembly configured for movement between a first actuated state and a first deactuated state, and wherein the drive assembly includes a pushbar operable to transition the drive assembly between the first actuated state and the first deactuated state; a latch control assembly configured for movement between a second actuated state and a second deactuated state; a lost motion connection operably connecting the drive assembly and the latch control assembly, wherein the lost motion connection is configured to move the latch control assembly from the second deactuated state to the second actuated state in response to movement of the drive assembly from the first deactuated state to the first actuated state, and wherein the lost

motion connection is further configured to permit the latch control assembly to remain in the second actuated state when the drive assembly moves from the first actuated state to the second deactuated state; and a coordination mechanism engaged with the latch control assembly and urging the latch control assembly toward the second actuated state with an actuating input force, and wherein the actuating input force is independent of the first actuated state and the first deactuated state.

2. The pushbar assembly of claim 1, further comprising a remote latching assembly operably connected with the latch control assembly, the remote latching assembly comprising: a first latch mechanism operably connected with the latch control assembly, wherein the first latch mechanism is operable to move from a first deactuated state toward a first actuated state in response to actuation of the latch control assembly; a second latch mechanism operably connected with the latch control assembly, wherein the second latch mechanism is operable to move from a second deactuated state toward a second actuated state in response to actuation of the latch control assembly; a first biasing member urging the first latch mechanism toward the first deactuated state such that the first latch mechanism exerts a first deactuating force on the latch control assembly; and a second biasing member urging the second latch mechanism toward the second deactuated state such that the second latch mechanism exerts a second deactuating force on the latch control assembly.

3. The pushbar assembly of claim 2, wherein the coordination mechanism includes a coordination mechanism biasing member urging the latch control assembly toward the latch control assembly second actuated state with the actuating input force.

4. The pushbar assembly of claim 2, wherein the pushbar has a projected position in the drive assembly first deactuated state and a depressed position in the drive assembly first actuated state; and wherein the drive assembly further comprises a return spring biasing the drive assembly toward the drive assembly first deactuated state.

5. The pushbar assembly of claim 2, wherein the first latch mechanism is operably connected with the latch control assembly via a first connector; wherein the first latch mechanism comprises a first housing and a first linkage movably mounted to the first housing, and wherein the first linkage has a first linkage deactuated position in the first latch mechanism deactuated state and a first linkage actuated position in the first latch mechanism actuated state; wherein the first biasing member is mounted to the first housing and biases the first linkage toward the first linkage deactuated position; wherein the first connector is connected with the first linkage and is configured to drive the first linkage toward the first linkage actuated position in response to actuation of the latch control assembly; wherein the second latch mechanism is positioned at a second location remote from the actuation assembly and is operably connected with the latch control assembly via a second connector; wherein the second latch mechanism comprises a second housing and a second linkage movably mounted to the second housing, and wherein the second linkage has a second linkage deactuated position in the second latch mechanism deactuated state and a second linkage actuated position in the second latch mechanism actuated state; wherein the second biasing member biases the second linkage toward the second linkage deactuated position; and wherein the second connector is connected with the second linkage and is configured to drive the second linkage toward the second linkage actuated position in response to actuation of the latch control assembly.

6. The pushbar assembly of claim 5, wherein the first latch mechanism further comprises a latchbolt configured for movement between a latching position and an unlatching position; wherein the first latch mechanism further comprises a blocking member operably connected with the first linkage, the blocking member having a blocking position in response to the first linkage deactuated position, and the blocking member having an unblocking position in response to the first linkage actuated position; wherein, with the blocking member in the blocking position, the blocking member retains the latchbolt in the latching position; wherein, with the blocking member in the unblocking position, the blocking member does not block movement of the latchbolt between the

latching position and the unlatching position; and wherein the second latch mechanism further comprises a deadbolt movably mounted to the second housing and engaged with the second linkage, the deadbolt having an extended position in response to the second linkage deactuated position, and the deadbolt having a retracted position in response to the second linkage actuated position.

7. The pushbar assembly of claim 6, wherein the remote latching assembly further comprises a retaining member operable to retain the first latch mechanism in the first actuated state by moving between a retaining position and a non-retaining position; and wherein the retaining member is configured to permit movement of the blocking member between the blocking position and the unblocking position when the latchbolt is in the latching position, and to retain the blocking member in the unblocking position when the latchbolt is in the unlatching position.

8. The pushbar assembly of claim 2, wherein the coordination mechanism further comprises an anchor bracket, and wherein the coordination mechanism includes a biasing member engaged between the anchor bracket and the latch control assembly.

9. The pushbar assembly of claim 1, further comprising a mounting assembly configured for mounting to a door; wherein the latch control assembly includes a movable component mounted for movement relative to the mounting assembly, the movable component having an actuated position in the second actuated state, and the movable component having a deactuated position in the second deactuated state; wherein the coordination mechanism comprises an anchor bracket and a biasing member; wherein the biasing member is engaged between the anchor bracket and the movable component and exerts an actuating force urging the movable component toward the actuated position, thereby contributing to the actuating input force; and wherein the anchor bracket is fixedly mounted to the mounting assembly and provides an anchor point for the actuating force exerted by the biasing member.

10. The pushbar assembly of claim 9, wherein the biasing member has a first end portion and an opposite second end portion, wherein the anchor bracket is engaged with the first end portion and limits movement of the first end portion in a first direction, and wherein the movable component is engaged with the second end portion and limits movement of the second end portion in a second direction opposite the first direction.

11. The pushbar assembly of claim 10, wherein the anchor bracket defines a channel and includes a flange projecting into the channel, wherein the biasing member is received in the channel, and wherein the flange is engaged with the first end portion.

12. The pushbar assembly of claim 1, wherein the latch control assembly is urged toward the second actuated state by a cumulative actuating force, and wherein the latch control assembly is urged toward the second deactuated state by a cumulative deactuating force; wherein the pushbar assembly further includes a remote latching assembly comprising: a first latch mechanism positioned remotely from the case, wherein the first latch mechanism is operably connected with the latch control assembly via a first connector such that actuation of the latch control assembly causes a corresponding actuation of the first latch mechanism, and wherein the first latch mechanism is configured to selectively exert a first deactuating force contributing to the cumulative deactuating force; and a second latch mechanism positioned remotely from the case, wherein the second latch mechanism is operably connected with the latch control assembly via a second connector such that actuation of the latch control assembly causes a corresponding actuation of the second latch mechanism, and wherein the second latch mechanism is configured to exert a second deactuating force contributing to the cumulative deactuating force; wherein the pushbar assembly has a first condition in which the first deactuating force contributes to the cumulative deactuating force, the cumulative deactuating force exceeds the cumulative actuating force, and the latch control assembly is biased to the second deactuated state; and wherein the pushbar assembly has a second condition in which the first deactuating force does not contribute to the cumulative deactuating force, the cumulative actuating force exceeds the cumulative deactuating force, and the

latch control assembly is biased to the second actuated state.

13. The pushbar assembly of claim 12, wherein the first latch mechanism comprises a first bolt having a first extended position and a first retracted position; wherein the second latch mechanism comprises a second bolt having a second extended position and a second retracted position; wherein the first latch mechanism is configured to retain the first bolt in the first extended position when the first latch mechanism is deactuated; and wherein the second latch mechanism is configured to retain the second bolt in the second extended position when the second latch mechanism is deactuated.

14. The pushbar assembly of claim 13, wherein the coordination mechanism is configured to exert the actuating input force contributing to the cumulative actuating force.

15. A system including the pushbar assembly of claim 12, the system further comprising a door having a first face, a second face opposite the first face, a top edge, a bottom edge opposite the top edge, a hinge edge, and a swinging edge opposite the hinge edge; wherein the pushbar assembly is mounted to the first face of the door; wherein the first latch mechanism is mounted to the door adjacent the top edge; wherein the second latch mechanism is mounted to the door adjacent the bottom edge; and wherein each of the first latch mechanism and the second latch mechanism is nearer to the swinging edge than to the hinge edge.

16. The system of claim 15, wherein the door further comprises a door preparation in which the remote latching assembly is positioned; wherein the door preparation comprises: an upper cavity extending downward from the top edge, wherein the first latch mechanism is seated in the upper cavity; an upper channel extending downward from the upper cavity, wherein the first connector extends through the upper channel; a lower cavity extending upward from the bottom edge, wherein the second latch mechanism is seated in the lower cavity; and a lower channel extending upward from the lower cavity, wherein the second connector extends through the lower channel; and wherein the free edge of the door is substantially unbroken by the door preparation.

17. A pushbar assembly, comprising: a drive assembly configured for movement between a first actuated state and a first deactuated state via actuation of a pushbar; a latch control assembly configured for movement between a second actuated state and a second deactuated state; a lost motion connection operably connecting the drive assembly and the latch control assembly, wherein the lost motion connection is configured to move the latch control assembly from the second deactuated state to the second actuated state in response to movement of the drive assembly from the first deactuated state to the first actuated state, and wherein the lost motion connection is further configured to permit the latch control assembly to remain in the second actuated state when the drive assembly moves from the first actuated state to the second deactuated state; and a coordination mechanism engaged with the latch control assembly and urging the latch control assembly toward the second actuated state with an actuating input force, wherein the actuating input force is independent of the first actuated state and the first deactuated state.

18. The pushbar assembly of claim 17, wherein the latch control assembly includes a movable component having an actuated position in the second actuated state, and the movable component having a deactuated position in the second deactuated state; and wherein the coordination mechanism comprises a biasing member engaged with the movable component and exerting an actuating force to urge the movable component toward the actuated position, thereby contributing to the actuating input force.

19. A pushbar assembly, comprising: a pushbar drive assembly configured for movement between a drive actuated state and a drive deactuated state; a latch control assembly configured for movement between a latch actuated state and a latch deactuated state; a lost motion connection operably connecting the drive assembly and the latch control assembly, wherein the lost motion connection is configured to move the latch control assembly from the latch deactuated state to the latch actuated state in response to movement of the drive assembly from the drive deactuated state to the drive actuated state, and wherein the lost motion connection is further configured to permit the

latch control assembly to remain in the latch actuated state when the drive assembly moves from the drive actuated state to the drive deactuated state; and a coordination mechanism engaged with the latch control assembly and urging the latch control assembly toward the latch actuated state with an actuating input force, wherein the actuating input force is independent of the drive actuated state and the drive deactuated state.

20. The pushbar assembly of claim 19, wherein the latch control assembly includes a movable component having an actuated position in the latch actuated state, and the movable component having a deactuated position in the latch deactuated state; and wherein the coordination mechanism comprises a biasing member engaged with the movable component and exerting an actuating force to urge the movable component toward the actuated position, thereby contributing to the actuating input force.
